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SMT. INDIRA GANDHI COLLEGE OF ENGINEERING
An Autonomous Institute with NAAC 'A' Grade
Approved by AICTE, Affiliated to the University of Mumbai

DEPARTMENT OF COMPUTER ENGINEERING

MAJOR PROJECT I (SEM VII) — AY 2026-27

Initial Project Document (IPD)

• Title of the Project :

Fine-Tuning Speech Note Transcription: A Real-Time Multi-Device Broadcast Transcription Tool for Classrooms and Meetings

• Problem Statement :

In classrooms and meeting rooms, students and participants spend much of their attention writing down what is said, which reduces engagement and often results in incomplete or inaccurate notes. Existing note-taking and captioning tools are typically single-device, single-language solutions that require every listener to run their own recording app, do not work reliably when the speaker switches between languages mid-sentence (e.g., Hinglish or other regional code-mixed speech), and provide no way to preserve non-verbal references such as a speaker pointing at a diagram or the board. There is a need for a system where one device records and transcribes the speaker in real time while every other device in the same physical room instantly and securely views the live transcript, with the ability to pause and resume the session and to flag moments where the speaker is referring to visual content that the transcript itself cannot capture.

• Objectives of the Project :

1. To develop a web-based application in which a single host device records and transcribes speech in real time, while other devices present in the same room join the session and view the live transcription simultaneously, without needing to record audio themselves.
2. To design a session-scoped broadcasting mechanism (e.g., room codes/QR-based session join over a local or authenticated channel) so that only devices physically present in that meeting/classroom session can view the live transcript.
3. To support real-time transcription with pause and resume controls, allowing the host to temporarily halt capture and continue the same session without losing prior transcript continuity.
4. To provide automatic speech recognition and translation for four languages — English, one international language, and two regional languages — including detection and handling of mixed-language (code-switched) speech such as Hinglish, and to normalize all recognized speech into a single unified English transcript.
5. To implement a contextual visual-reference tagging feature that detects when a speaker is verbally referring to non-audible content (e.g., "as shown in this diagram", "look at the board") and inserts an inline tag/marker in the transcript at that point, even though the visual content itself is not captured.
6. To fine-tune the underlying speech-to-text model on domain-specific classroom/meeting vocabulary and code-mixed regional speech patterns to improve transcription accuracy over generic off-the-shelf ASR.
7. To generate a downloadable, structured transcript/notes file at the end of each session, preserving language-normalized text, tags for visual references, and pause/resume segment boundaries.

- **Scope of the project :**

The system is scoped to in-person classroom and meeting-room scenarios where a single host device performs speech capture, recognition, and translation, while any number of other devices in the same room connect to a shared session to view the live transcript in real time. It covers real-time multilingual and code-mixed automatic speech recognition, session-based multi-device broadcasting restricted to the same room/session, pause-resume session control, contextual tagging of visual references, and end-of-session downloadable notes generation. The scope does not extend to remote/multi-room participation, video-conferencing platform integration, or actual capture/recognition of the visual content itself (diagrams, slides, whiteboard) — such moments are only flagged as tags, not transcribed visually, in the current phase.

- **Challenges in the project & projected solution**

1. Broadcasting a live transcript from one host device to multiple listener devices with low latency while keeping the session restricted to the room — Solution: use a WebSocket/real-time streaming architecture with a session token or QR-based room code, so only devices that join the specific session receive the live transcript feed.
2. Recognizing and normalizing mixed-language, code-switched speech (e.g., Hinglish, English mixed with a regional language) into a single coherent English transcript without manual language selection — Solution: use multilingual ASR/LLM-based language identification per utterance combined with fine-tuning on code-mixed regional datasets, followed by machine translation/normalization into English.
3. Detecting that a speaker is referring to visual content not present in the audio (e.g., diagrams, gestures towards the board) — Solution: use keyword and intent-pattern detection (e.g., deictic phrases like "this diagram", "as shown here") on the live transcript stream to trigger an inline visual-reference tag at that timestamp.
4. Maintaining transcript continuity and synchronization across pause/resume actions and multiple simultaneously connected listener devices — Solution: maintain a server-side session state machine with segment timestamps, so resuming after a pause appends to the same session transcript and all connected devices re-sync automatically.
5. Achieving acceptable accuracy for regional and code-mixed languages with limited labelled data — Solution: fine-tune a pretrained multilingual speech model (e.g., Whisper/IndicWhisper-class models) using LoRA/PEFT on available code-mixed and regional-language datasets rather than training from scratch.

- **Input :**

Live speech captured from a single host device's microphone in a classroom or meeting room, in English, one international language, two regional languages, or any mix thereof; host controls for starting, pausing, and resuming the session.

- **Process :**

Automatic Speech Recognition (ASR) fine-tuned for classroom/meeting and code-mixed regional speech, Language Identification for code-switched utterances, Machine Translation/normalization into unified English text, real-time session broadcasting over WebSockets to all connected room devices, NLP-based deictic/reference detection for visual-content tagging, and session-state management for pause/resume continuity.

- **Output :**

A live, unified-English transcript broadcast in real time to every device connected to the session; inline tags marking moments where the speaker refers to a diagram, board, or other visual content; a pausable/resumable live session view; and a downloadable, structured transcript/notes file once the session ends.

- **Hardware / Software Platform used :**

Hardware: A standard laptop/PC/tablet acting as the host device with a built-in or external microphone; listener devices (phones/laptops/tablets) with a web browser and no special hardware; a cloud or on-premise server (GPU-optional) for running the fine-tuned ASR model.

Software: React.js-based web front end; Python/Node.js backend; WebSocket-based real-time broadcasting layer; fine-tuned multilingual ASR model (e.g., Whisper/IndicWhisper-class models with LoRA/PEFT fine-tuning); NLP/LLM APIs for translation, normalization, and reference-tag detection; MongoDB/PostgreSQL database; QR-code-generation and notes/PDF export libraries.

- **IEEE Base Reference Papers :**

1. Using Speech Recognition for Real-Time Captioning and Lecture Transcription in the Classroom — https://ieeexplore.ieee.org/document/6529071/
2. Code-Switching ASR for Low-Resource Indic Languages: A Hindi-Marathi Case Study — https://ieeexplore.ieee.org/document/10835062/
3. Multilingual Automatic Speech Recognition for Indian Languages — E2E Framework — https://ieeexplore.ieee.org/abstract/document/10973636/

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